

A Lightweight LLM-Driven Multi-Agent Framework for Virtual Society Simulation with Memory Feedback

Aoao Guo · Xinyi Tao · Huakang Li · Xuewu Dai

University College London | Xi'an Jiaotong-Liverpool University | Northumbria University

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Valentown — seven residents who plan, meet, remember and adapt

Two ways to simulate people

R Rule-based agent models

- The standard tool for social networks, transport, smart cities and large socio-technical systems
- Agents follow predefined rules or handcrafted policies
- Little contextual reasoning, no long-term planning, no memory
- Strong on macro patterns, weak on human-like individual behaviour

L LLM-driven agents

- Generate coherent language, hold context, imitate aspects of reasoning
- Push simulation past rigid rules toward expressive, flexible behaviour
- But most systems act reactively — on the immediate prompt only
- Interaction is often ad hoc or purely stochastic

The gap: no explicit structure for behaviour, interaction and feedback. Over a long run behaviour drifts and becomes inconsistent, emergent dynamics are hard to control, and results are hard to reproduce or analyse systematically.

One question, three requirements

How can we design a virtual-society simulation in which agents behave in a structured, evolving and controllable way over time?

1 Temporal structure

Behaviour has to be organised as a day, not as a stream of one-off reactions.

OUR ANSWER

Schedule-driven agent design

2 Behavioural evolution

What happened yesterday has to change what an agent does tomorrow.

OUR ANSWER

Memory-based feedback loop

3 Controllability

Interaction must follow an explicit rule, and a human must be able to step in.

OUR ANSWER

Co-location trigger + human-in-the-loop

A backend that thinks, a frontend that shows

B Backend — Python

- Agent cognition: daily planning, dialogue, reflective reasoning
- Memory management and all calls to the LLM (DeepSeek API)
- Modular by design, so parts can be replaced independently

F Frontend — Phaser

- Renders the town, agent movement and interactions
- Pulls agent states, plans and events over a REST API

FIVE IMPLEMENTATION MODULES

Agent Manager · Scheduler · Interaction Engine · LLM Interface · Memory Manager

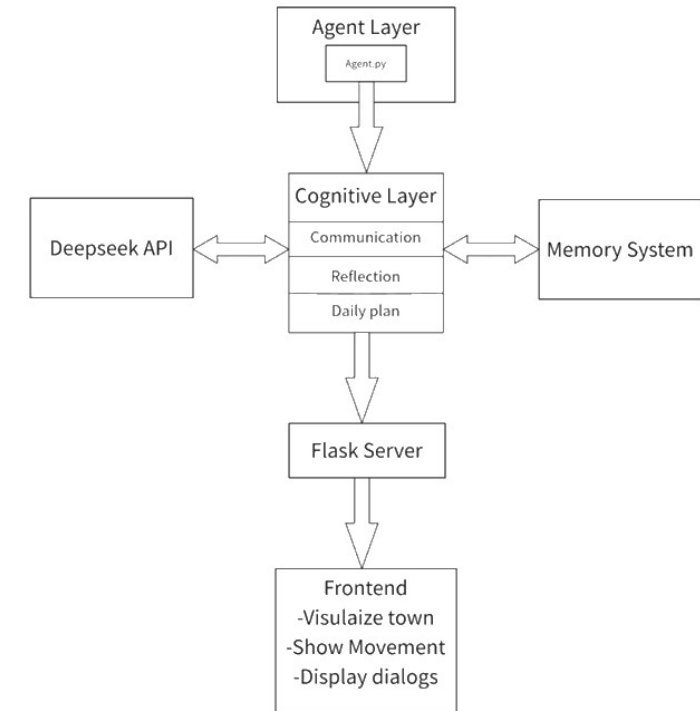


Fig. 1 Overview of the proposed framework

Plan the day, then let the town decide who meets

1 Schedule-driven behaviour

- Each day begins with an explicit schedule, not a reactive step
- Prompt = identity, personality, goals + memories retrieved for context
- The LLM returns structured activities and their implied destinations
- Routines become temporally coherent, and readable to us

Morning · home

Midday · café

Afternoon · park

Evening · home

2 Spatio-temporal interaction

- A conversation fires when two agents share a location in the same period
- The trigger is deterministic — controllable, repeatable, easy to analyse
- What is said is generated by the LLM from context and recent history
- Interaction is grounded in the shared environment, not sampled at random

same place + same period → they talk

Deterministic when, generative what — the framework keeps control of the dynamics while the language model supplies the diversity.

Memory is where the days connect

3 Memory-based feedback loop

- Plans are stored as behavioural records, conversations as social experience, reflections as high-level summaries
- Every entry carries a category, a timestamp and an importance score
- Retrieval favours recent and high-importance entries; the store stays bounded by prioritisation
- Reflection compresses raw experience into insight, written back for the next planning cycle

OPTIONAL: HUMAN IN THE LOOP

A person can write feedback straight into an agent's memory — the same channel the agent's own experience uses. Extra controllability, never required for baseline operation.

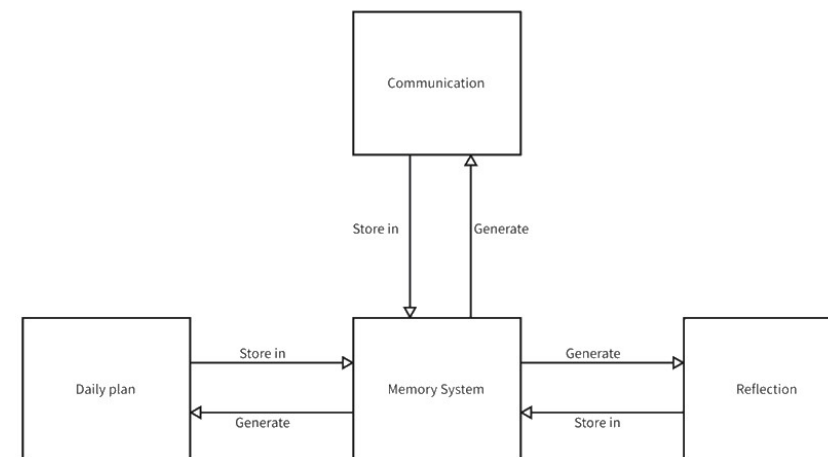


Fig. 3 Memory-centred feedback loop

The language model itself stays stateless. Every bit of continuity lives in the memory system — which is exactly what makes agent state inspectable and replaceable.

One simulation day, end to end

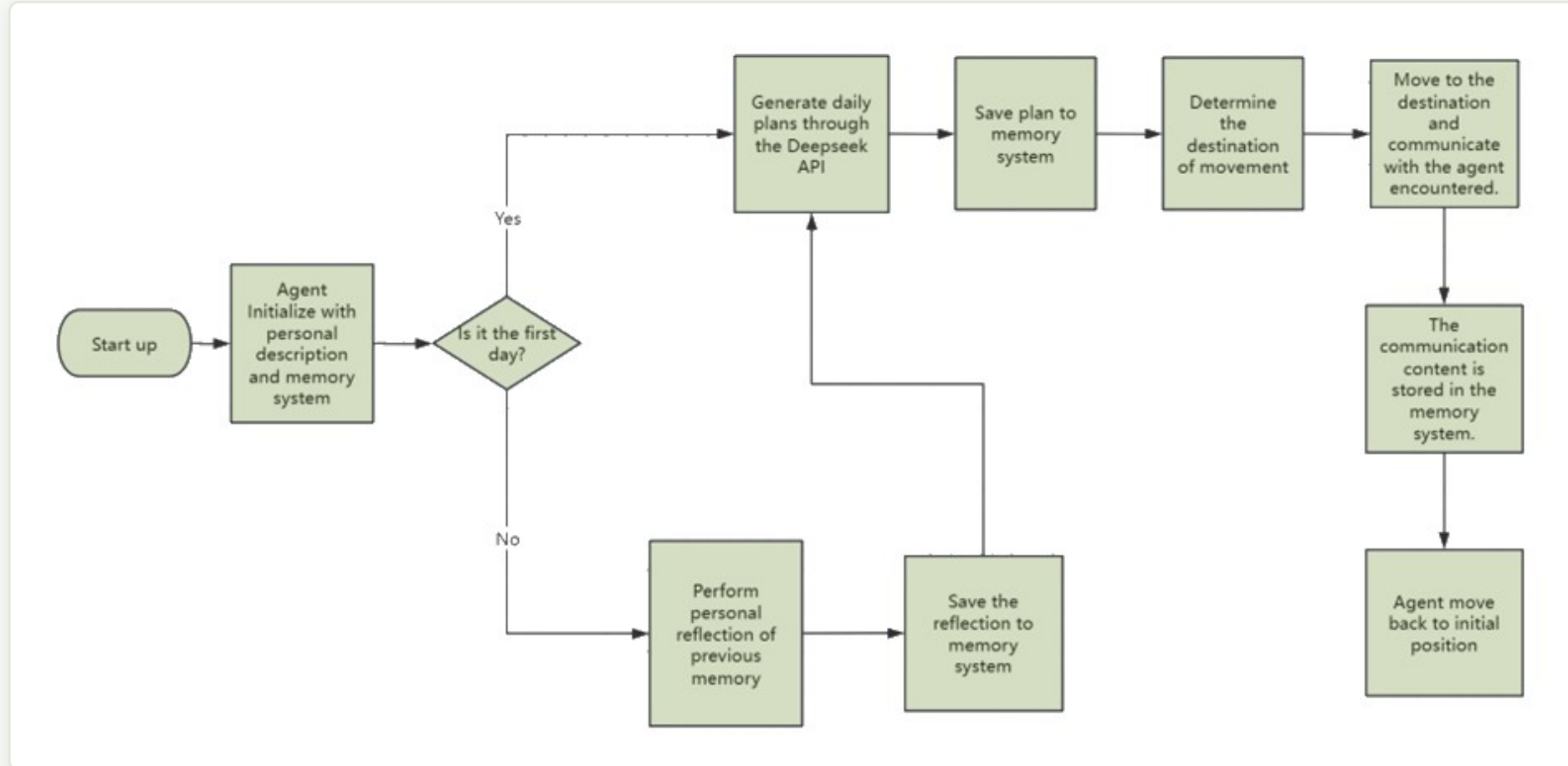


Fig. 2 · Day 1: agents plan straight from their profile. Every later day: retrieve and reflect on memory first, then plan — the loop is closed.

Valentown: the framework, running



The running client: the whole town on one plane, each resident's current action and internal state, and the day's conversation log.

Valentown

The case-study town: homes, supermarket, pharmacy, café and park as spatial anchors

7 residents

Heterogeneous age, role, personality, goals and description — the framework is not tied to them

DeepSeek API

One interface for schedules, dialogue and reflection: prompt construction, calls, response validation

Python + Phaser

Backend pre-computes behaviour asynchronously; the client replays it smoothly over REST

What we ran, and what we looked for



Ron Parker

60 · supermarket owner



Ella Parker

58 · pharmacy owner



Emma Harris

30 · mother



Gavin Harris

32 · father



Adam Harris

7 · child



Mia Thompson

28 · family teacher



Arthur Morgan

29 · architect

Environment

A structured town of predefined locations. Agents start at fixed home positions and move to plan destinations; co-location is what enables interaction.

Duration

Multiple consecutive virtual days. Each day is a full cycle: schedule, movement, interaction, memory logging, reflection.

Execution

Scheduling and interaction triggers are deterministic; LLM content is stochastic. Reproducible pipeline, diverse behaviour.

Evaluation focus: what interaction does to behavioural diversity · what memory and reflection do over time · whether social structure emerges from simple rules

RESULT 1

Turn interaction off, and the town goes quiet

INTERACTION ENABLED

- Noticeably richer and more diverse activities and social engagements
- Conversations inject new context into memory, which then shapes the next plan
- Agents adjust future plans on what was said — follow-up meetings, revisited locations
- Repeated contact reinforces relationships between specific pairs

INTERACTION DISABLED

- Behaviour becomes repetitive and isolated
- No new social information enters the system at all
- Agents fall back on their initial goals and their previous plans
- Schedules stay largely static from one day to the next

Interaction is what keeps the routines moving: it drives behavioural diversity and day-to-day adaptation at the same time.

RESULT 2

Memory changes shape as the days pass

EARLY DAYS

Memory is dominated by daily plans and basic observations

AS DAYS PASS

Communication events and reflection outputs become increasingly prominent

LATER

High-importance interaction and reflection memories form the dominant context for new decisions

And memory content tracks what agents do next

- Agents with rich interaction and reflection memories behave more adaptively and more coherently in social terms
- Repeated interaction between specific agents forms persistent bonds — visible later as repeated co-location and coordinated activity
- Reflection-derived insight pushes agents toward goal-aligned behaviour, so task selection observably shifts across days
- Where memory influence is limited, adaptation is weaker and behaviour less consistent over time

Memory is the mechanism that turns yesterday's experience into today's plan — and simple generative rules are enough to produce social structure nobody hand-coded.

What this buys us, and what it costs

Advantages

- 1 Lightweight & reproducible**
Clearly defined modules, an explicit pipeline, and deterministic spatio-temporal triggers
- 2 Controllable**
Schedule-driven behaviour, explicit co-location rules, optional human feedback
- 3 Extendable**
Agent Manager, Scheduler, Interaction Engine, Memory Manager and LLM Interface swap independently

Limitations

- 1 Cost**
LLM calls carry computational and financial cost that grows with agents and simulation days
- 2 Scale**
The study is small — few agents, a simple town; larger and more complex settings still need validation
- 3 Stochasticity**
Generated output varies between runs, which makes strict quantitative evaluation harder

Where this leaves us

What we showed

- A schedule-driven multi-agent framework where planning, interaction and reflection are linked through a memory-centred feedback loop
- Importance-based memory preserves what mattered; reflection turns it into insight that guides the next day
- Simple generative mechanisms produce consistent routines, lasting social relationships and experience-driven adaptation — with no hand-designed rules
- Memory-driven LLM agents are a practical and interpretable foundation for scalable virtual-society simulation

Future work

1 Quantitative evaluation

Social network analysis, interaction frequency, plan coherence, task completion — plus user studies on perceived realism

2 Scale & real time

Larger populations and richer environments; moving from pre-generated behaviour to real-time reaction to events

3 Reasoning & applications

Multi-level reflection and evolving goals; game NPCs, educational simulation, smart environments, embodied agents

Thank you

Questions are very welcome.

Aoao Guo

Faculty of Life Sciences, University College London
aoao.guo.25@ucl.ac.uk

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